

20260515_Paper_Directions_11

📅 Fri, May 15, 2026 3:31PM ⌚ 37:45

SUMMARY KEYWORDS

mean pool ability, quantile bins, team ability, heavy-tailed distribution, player assignment, standard deviation, local rank, assortative nature, uniform distribution, beta distribution, epsilon noise, team performance, probability of success, nonlinear effect, algorithm development

SPEAKERS

Speaker 2, Speaker 1, Charles Levine, Alex Gates

A Alex Gates 00:13
Here's what I'm thinking, right, our primary x axis is the mean

S Speaker 1 00:28
pool

A Alex Gates 00:30
ability,

C Charles Levine 00:32
that's where our bins came from,

A Alex Gates 00:35
minus me quantum I

C Charles Levine 00:41
right, that's what that's what served us well for the other,

A Alex Gates 00:44
for the other,

C Charles Levine 00:45
for the other domains,

A Alex Gates 00:46
right? Yeah, so you're now implementing this as your x axis for all simulations and everything,
right? Okay, so then the next piece

S Speaker 1 00:57
is Is

A Alex Gates 01:12
if we think about all three systems, what is the distribution looking like of this?

C Charles Levine 01:21
Okay, of that mean

A Alex Gates 01:23
of that mean of the the mean pool ability minus me right for

C Charles Levine 01:30
with the

A Alex Gates 01:31
with the army it was like this right

C Charles Levine 01:33
now at mean pool ability,

A Alex Gates 01:36
yeah. If

C Charles Levine 01:48
I'm not mistaken, this was what it looked like for the individuals.

A Alex Gates 02:00
Huh,

C Charles Levine 02:04
that is what it looked like for the pools.

A Alex Gates 02:09
Oh, so it was pretty flat.

C Charles Levine 02:10
Yeah, it was. Yeah, it looked like it was not

A Alex Gates 02:15
great. Okay, so this is really like what it looks like in any of the three data sets, then then I'm much happier, then I'm not as worried, but that was one of the first, first things that I was kind of playing around with, so then the second thing, and I mentioned this in an email to you, is that I was worried that the number of teams that you were actually using was approximately close to the number of quantile bins, right, and really in every scenario that we've looked at, this should be way bigger than the number of quantile bins,

C Charles Levine 02:55
right.

A Alex Gates 02:55
So one easy thing I thought just up in the simulations is your number of bins compared to the number of quantiles, right? And make this really, I mean, the number of teams,

C Charles Levine 03:07
teams,

A Alex Gates 03:07
pools, whatever you want, yeah, right, is being much larger. And so then in our simulated environment space, right, you had all of the teams were kind of very well structured, right, and they didn't, I mean, the team range was defined, but then you added, you know, the players had a plus epsilon noise

C Charles Levine 03:34
right

A

Alex Gates 03:35

happening, right? I think what you need to kind of start is not a range for your teams, but a like, what's the right way to kind of do that? Maybe you need to draw a team ability, which is treated like the mean and Mean, mean ability. Let's see the target mean ability of the team of the pool or team, whatever. I guess I gotta work my language, yeah, right. And then you, you have all of these there, and then the players can be assigned to their target teams,

C

Charles Levine 04:30

right? So this is exactly what I was thinking about this morning.

A

Alex Gates 04:36

Yeah,

C

Charles Levine 04:36

let me see if I can back brief slash tell you how I would say it. So in the beginning we said we'll either go, we'll either draw from a uniform distribution or a beta two or some sort of a normal distribution, right? Or, but we'll draw random abilities for all the players, then we'll line them all up from one to n based on that, right. It the naive way is to then just create bend them by creating what's the word I'm looking for, it starts with a D

A

Alex Gates 05:12

digitized

C

Charles Levine 05:12

comprise, forgot what we said, but that's the cleanest way.

A

Alex Gates 05:17

Yep,

C Charles Levine 05:18

and we said, okay, well, that's not great. So then we said, well, let's add this epsilon, right? So what that does is it just kind of rearranges them, right?

A Alex Gates 05:28

Yeah,

C Charles Levine 05:29

well, then I thought that's not reality, right? Reality is Iowa State Bible College has the next Ralph Sampson on it once in a blue moon, and these, in other words, it's just, it's, it's heavy tailed, right, because we know that scale-free distributions are not rare.

A Alex Gates 05:48

Yep,

C Charles Levine 05:49

and so then I said, why don't we do this, why don't we take these people, why don't we create the bins first, right? Or, and I'm just thinking, I mean, it's something might sound stupid, but what we need to do is take some sort of heavy-tailed exponential distribution, and from that we create the epsilon, you see, so that some people who are awesome will end up down here, but most of the people who are awesome will end up here. Does that make sense.

A Alex Gates 06:21

Yeah,

C Charles Levine 06:22

and you're probably, you might be saying the same thing. There's different ways to skin a cat. I just want to make sure we're on the same page.

A

Alex Gates 06:28

Yeah, so, so I don't know if the two ways are equivalent. I guess I'm starting to move to a new shape, and that's

C

Charles Levine 06:52

I'm happy to go with your intuition. I think you might have one more belt.

A

Alex Gates 06:55

I don't know, no, for this I don't have an intuition for this. I'm riding with you. We're trying to figure it out. Okay, so, so this is my one problem. I like this general setup, but what this induces is actually this artificial hard difference between the teams, even if we've smudged them and done something,

C

Charles Levine 07:19

but we've already designated that the difference between teams exists,

A

Alex Gates 07:22

right, and I don't know if that's the right way of doing so. There's two other, there's actually two other different ways I can think of to create the teams.

C

Charles Levine 07:37

Okay,

A

Alex Gates 07:38

but I think that this mechanism of how to let's assign,

C Charles Levine 07:46
I think that's the key here. I think we're somehow, we're that's that pre sorting,

A Alex Gates 07:56
yep,

C Charles Levine 07:56
right, which we know is definitely of an end of an assortative nature,

A Alex Gates 08:01
yep,

C Charles Levine 08:01
in all three domains, it just turns out like that, right? So, I think recreating that pre-sorting, I agree, is our, is our primary challenge right now, I think.

A Alex Gates 08:10
Yeah, so, so, how do we do this is what we need to kind of readdress. Okay, one thing to look at that I was getting towards, but I'll just kind of get it more directly. Is that here I'm going to get a team ability or a pool ability, or whatever you want, and I'm going to pre-select that randomly, and then a player. Well, what letters, what do you use to represent a player or an officer?

S Speaker 2 08:45
I,

A Alex Gates 08:45
I, okay,

C Charles Levine 08:46
and so a like a sub I is that guy's ability,

A Alex Gates 08:49
yeah. So I goes into team J proportional to the distance between alpha I and and or so Alpha Team J

C Charles Levine 09:07
Cap A is the ability

A Alex Gates 09:09
Cafe great right so now we're centering on a team but instead of having, like, a hard bin cut off sorting type of thing, you could make this, for example, a Gaussian around T J, so that it tends to get people that look like that, or you could do something that's even more heavy-tailed, so that you get people that look like that.

C Charles Levine 09:40
Let me internalize this real quick. So, you're saying the probability that I goes to team J

A Alex Gates 09:45
is going

C Charles Levine 09:45
to be proportional to the that individual players performance metric minus, I guess, the mean of a team's performance metric. How do we create those mean team performance? These metrics,

A Alex Gates 10:00
you selected them first, so instead of creating hard bins, like sorted,

C Charles Levine 10:05
yeah,

A Alex Gates 10:05
you select now each bin, each pool, each team, whatever has its own,

C Charles Levine 10:14
but it's from a uniform distribution, so that's a true much more represents you're definitely going to have people on the right, you're going to have people on the left, but how close they are together is a whole different ball game.

A Alex Gates 10:28
So that's one way of doing it.

C Charles Levine 10:29
Okay,

A Alex Gates 10:39
another way of doing it is with, like, a preferential treatment type of thing, so, like, you would add in people, and at first the probability of a person going to a team is completely even probability, but then once the team starts to have people, you let

C Charles Levine 11:03
you recalculate,

A Alex Gates 11:04
you recalculate the time, and every time that's right.

C Charles Levine 11:08
Also, Barabasha-ish,

A Alex Gates 11:10
yeah,

C Charles Levine 11:10
right,

A Alex Gates 11:11
yeah,

C Charles Levine 11:12
rich get richer.

A Alex Gates 11:14
But the point here is to think through a little bit more how we're assigning players to teams, and this very strict sorting thing, so that the what I don't like about is that the teams are forced to be very ordered.

C Charles Levine 11:30
What if we,

A

Alex Gates 11:31

so what I want is something where there's some teams that look like this, some teams that look like that, some teams that have, you know, that, and then they overlap in many ways, like a bunch of teams have overlapping windows, like there's just this general Maloo of team of the way that the

C

Charles Levine 11:54

team, you say overlapping windows, you mean windows of

A

Alex Gates 11:56

talent, yeah, right, right, which kind of happens in the noise version, so it's not like that noise version, but I think the

C

Charles Levine 12:04

noise version is too linear, right? Right, it's definitely not getting us the outliers, and it's

A

Alex Gates 12:09

yeah,

C

Charles Levine 12:10

but more of a lattice. This is

A

Alex Gates 12:13

really challenging because you still have to have that knob of how assortative is the mixing, right? Right, so this kind of still will get you, because you could, you could change the standard deviation of the allowed probability, and and that might get you something that's much more concentrated versus spread out, but I think that the two things that we're missing in how we're structuring our teams is one, the number of teams has to be much larger than the number of bids that you're quantile,

 Speaker 2 12:50
right?

 Alex Gates 12:50
Right, so that's

 Charles Levine 12:51
clearly,

 Alex Gates 12:52
yeah, and

 Charles Levine 12:53
then otherwise it's meaningless,

 Alex Gates 12:54
yep,

 Charles Levine 12:54
right,

 Alex Gates 12:55
and the way that we're structuring the teams can't be very rigid, so we got to have a little bit more flexibility in how the teams are being created, sorted out,

C Charles Levine 13:04
so

A Alex Gates 13:05
right. So the first thing I want you to think about is this mechanism. Well, here's, here's a few like ways of doing it.

C Charles Levine 13:10
Would it be no mistake to just look at existing data and look at the teams historically and get a feel for where the means usually fall? What did center just did? distribution is. I mean, I have, I have historical data for all that. Would that help me to build the model, or would I be better off not just extra time?

A Alex Gates 13:33
Yes, I don't think it's.. it would be misleading to look at that,

C Charles Levine 13:39
but then again, it's not necessarily going to be something generalizable. We're building it from scratch. We want a multi-domain

A Alex Gates 13:45
correct

C Charles Levine 13:46
idea.

A

Alex Gates 13:47

So, where helps me is thinking a little bit about how much structure, like inequality and structure, is there present, and if it's present in all three, then that's something that we need to think about in the model. If it's present in only one of the three data sets that you've looked at, then that tells me it's not important, as like a one off thing, right?

C

Charles Levine 14:14

So I think we're going to find completely non partitioned talent intervals overlapping talent intervals in all three sets, right.

A

Alex Gates 14:26

Yep,

C

Charles Levine 14:27

and I think we're also going to find, I think we're going to find that in all three, that the overlapping talent sets, I think we're going to find the mean variation we We would need to pull not just the means but the standard deviations,

A

Alex Gates 14:49

so that's that's a question,

S

Speaker 2 14:51

right?

A

Alex Gates 14:51

Yeah, do you need to pull yet? Probably to make it, but I don't know if that's yeah. I don't know.

C Charles Levine 15:02
This is so cool.

A Alex Gates 15:03
Yeah, so, but this is the first mechanism I think we can improve, right? And I don't know if this is really going to fix the problem or not, but

C Charles Levine 15:12
I have looked at that machine I built,

A Alex Gates 15:14
yeah,

C Charles Levine 15:15
for hours and run all kind of stuff, and I can tell you that this is exactly the problem,

A Alex Gates 15:22
okay?

C Charles Levine 15:23
I'm certain that the assortativeity, the assorted, this, the pre-sorting is the

A Alex Gates 15:30
key. Yep. So, so this is the first thing. Then the second thing is, they

C Charles Levine 15:35
make sure I understand this correctly.

A Alex Gates 15:36
Yeah,

C Charles Levine 15:37
so what you're saying is, what we, the, yeah, the, the on compartment, the clean compartmentalization is not the way to go. So, what we first need to do is say, hey, let's, if we've got 100 teams, we need to pull 100 numbers between zero, and we need to get a uniform distribution for the average for the 100 or the average performance metric of each of those teams, and build the teams that way, and in order to have them be potentially overlapping, of course, then we probably need to build some sort of variation, which is probably more narrowly focused, right, and then we establish these teams and we lay them on the line on a number line, and then we say, okay, now let's go get these players, and let's, that we've given talent either through a uniform or through whatever distribution, we've given a talent through some distribution, and then let's say, okay, now we need to assign them to those teams, and we're going to assign them to those teams that we just built based on the latent ability diff distance from that team's mean, correct. Yeah, and perhaps we're going to have to, we're going to use an epsilon as well.

A Alex Gates 16:56
Yep,

C Charles Levine 16:57
right. Or just the probability of going there could be,

A Alex Gates 17:01
could be enough,

C Charles Levine 17:01
could be enough, but am I right so far?

A Alex Gates 17:04
Yes.

C Charles Levine 17:05
Okay. Good. Now we're saying part two.

A Alex Gates 17:09
Yeah, so the first thing is, how do we build teams? Right, let's call this one how we build teams, right? right, two is how we use local rank, so the other thing that I'm still trying to wrap my head around is even when they're in these teams and they're doing these things right, what makes it harder to get out of a tie team than a middle team?

C Charles Levine 17:58
What makes it harder to succeed? Yeah,

S Speaker 2 18:00
high team, that's the middle team,

A Alex Gates 18:02
exactly right, like there's there's got to be something, and so far all of the mechanisms that we've had actually in place says that it's about local rank, but local rank ends up being the same thing across all the teams, and so

C Charles Levine 18:17
local rank is broken down, it's broken down into the pros and the cons,

A Alex Gates 18:21
yeah,

C Charles Levine 18:22
right, and the pros of development, etc. but the cons are, and this is really where distinction comes in, the cons are the ability to, when you, when you're down here, up here, there's, there's only so much, everybody's so close together, talent wise,

A Alex Gates 18:41
yep,

C Charles Levine 18:41
right, and it's also high that it's not just the fact that it's difficult for the rater, but it's also difficult for the global person to differentiate between the people at that high,

A Alex Gates 18:55
so that's the question that's I think that

C Charles Levine 19:00
works for basketball, and it works for the army. It's not

A Alex Gates 19:03
going to work

C Charles Levine 19:03
for academics, because that's purely a purely local decision, whether who they continue to.

A Alex Gates 19:12
Yeah, so here's one question. Maybe this is worth going into the data for. You've only looked at the mean pool ability. What about the standard deviation of the pool ability? What if you did a heat map where you have your mean down here and you can quantile it, and then you also have your standard deviation here,

C Charles Levine 19:42
I It here as a y axis,

A Alex Gates 19:44
as a y axis, and then you shade it based on the probability,

C Charles Levine 19:53
probability of

A Alex Gates 19:54
promotion, whatever. Yeah, so. So success based on probability of success, right.

C Charles Levine 20:02
Okay, so tell me again, what the x axis is.

A Alex Gates 20:05
So an x axis is the what you've had before, the mean pool.

C Charles Levine 20:08
So this is for pools,

A Alex Gates 20:10
this is for pools,

C Charles Levine 20:10
that's the pool mean,

A Alex Gates 20:11
yep,

C Charles Levine 20:12
the pool mean talent performance metric,

A Alex Gates 20:15
correct. And then this is the standard deviation

C Charles Levine 20:18
of that talent of that pool

A Alex Gates 20:19
of that pool

C Charles Levine 20:20
talent,

A Alex Gates 20:20
so the hypothesis is that if the standard deviation is really large, then everybody's differentiated,

C Charles Levine 20:35
and there's not as much of a crowding effect,

A Alex Gates 20:37
right,

C Charles Levine 20:38
and there's also okay,

A Alex Gates 20:40
low crowding,

C Charles Levine 20:43
because the other thing I was gonna say, for the con is the opportunities, opportunity for command, opportunity for first author, opportunity to be on the court.

A Alex Gates 20:52
If the standard deviation is low, then that's high crowding,

C Charles Levine 20:57
right?

A Alex Gates 20:59
The effect should be stronger here.

C Charles Levine 21:01
Distinction should be easy with high standard deviation.

A Alex Gates 21:05
Distinction should be easy with high standard deviation. Yeah, as standard deviation goes up, crowding is lower. Distinction should be easier. Low standard deviation is high crowding, distinction should be harder. We should see the stronger nonlinearity here, right? So then basically the idea is nonlinear, u is stronger,

S Speaker 2 21:29
 u is weaker,

A Alex Gates 21:33
here. Okay, so this is the first thing to kind of check in the data and data sources, but then if this is really what's holding up. Then we need to do something that affects this more, so what mechanism?

C Charles Levine 21:46
The standard deviation.

A Alex Gates 21:47
Yeah, what mechanism can we use to kind of tweak that

C Charles Levine 21:53
standard deviation?

A Alex Gates 21:54
Yeah,

C Charles Levine 21:55
I'm saying that because I'm recording it. I wanted to share.

A Alex Gates 21:57
Yes. Sorry. So that's that's a second where I think that by ranking we lose this right. So ranking does not do standard deviation, ranking is just pure ranking.

C Charles Levine 22:12
Yeah,

A Alex Gates 22:13
so it's independent of how much crowding there is. Crowding is the same, ranks are the same regardless of crowding, and so that's the it was an easy thing to use, the

C Charles Levine 22:25
median versus the mean,

A

Alex Gates 22:26

right? It was an easy thing to use, but now I'm questioning if that, if that's missing what's going on. So, here are two different things, and I actually don't know which one is the right one to really worry about the most, but here are two different things that we could question about the setup that we have, and if you change the mechanisms a little bit, might help to give us what I

C

Charles Levine 22:50

think. Absolutely, going after the team building first is the most

A

Alex Gates 22:55

likely,

C

Charles Levine 22:55

not just most likely, but we're looking for a minimal model,

A

Alex Gates 23:01

right,

C

Charles Levine 23:02

and this already is - we're already starting to, I mean, it might be explored. The chances that that graph, the heat map is going to be the same for all three domains. I don't know how strong that is. Whereas down here we might be able to make a change for how we build teams that will be ubiquitous,

A

Alex Gates 23:22

yep,

C Charles Levine 23:22
and give us good results,

A Alex Gates 23:24
yep,

C Charles Levine 23:24
and that's another paper,

A Alex Gates 23:25
right? That, that could be..

C Charles Levine 23:32
I want to make sure I understand this. We're talking, you're talking about, you're saying, as an exploratory analysis, let's create this plot where we go, where the x axis is the pool's mean talent and the y axis is the standard deviation of that pool's talent.

A Alex Gates 23:49
Yeah,

C Charles Levine 23:50
and then we're going to color it based on how many players in that pool were selected,

A

Alex Gates 23:58

right? So, for the plot you've done beforehand, this wasn't there, right? And you did mean pool to this, right? And what you had is the quantile bins, and it was darkest here, and then it got a little lighter, very light, right? And it got a little lighter and very light, right? So so that's that was the non, that's the U shape that you found.

C

Charles Levine 24:27

What's unfortunate is there's only 60, there's only 60 successes per year.

A

Alex Gates 24:34

Oh, so do it in the army, do it in whatever, all three, all three of our systems to me are exchangeable,

C

Charles Levine 24:42

yeah, and the code is too,

A

Alex Gates 24:44

yeah, of course, I know you don't want to go back to no, I just say that at the moment I don't mind,

C

Charles Levine 24:50

I mean,

A

Alex Gates 24:50

of course I isn't right, so the third thing that's in my mind that we still have. A satisfactory integrated in was when you went in and you looked at high ability, middle ability, and low ability, and you showed that actually you don't get the, you really, you get the

C Charles Levine 25:25
pretty much monotonically decreasing.

A Alex Gates 25:27
It was monotonically there, it kind of came up and then went down, and then for high, what was it?

C Charles Levine 25:33
Same monotonically decreasing,

A Alex Gates 25:35
monotonically decreasing,

C Charles Levine 25:37
because in no sense, right?

A Alex Gates 25:38
Yeah, so I don't think anything we've done gets at this at all.

C Charles Levine 25:49
Okay, I'm trying to think of where to put that on the Christmas tree. I'm trying to think in my head of how I integrate that into what we're talking about, do Yes, but we should still be able to get a composite graph that mimics that. I mean, what, that's to me, that isn't really news, because all of our stuff has a little bit of a little bit of this in the front before it does that, and that I saw this, I think I've seen this almost, and it's like this doesn't make sense, but that probably has something to do with attrition.

A Alex Gates 26:29
Yeah,

S Speaker 1 26:30
okay,

C Charles Levine 26:31
you know what I'm saying, but, but I mean, that this is just this reinforces the composite of this, doesn't it?

A Alex Gates 26:39
Yeah, but this might be the thing that we should be going after to create the composite and not the composite by itself, right. So, what this is saying, right, is that actually, if you have three lines of different slopes, or whatever it is, right, then the adding of those three lines might get you something, but it's not. It shouldn't be monotonic, because this is definitely.. I just

C Charles Levine 27:06
know that the top one was monotonic. I remember I looked at it, and I was like, it's not just that it drops off, it starts high and strictly goes down, and I make them.. of course that's a function of where do you, where do you pick the mid, I just divided it equally into thirds in terms

A Alex Gates 27:27
of try to pull up that figure again, I want to know what the middle does, okay, is it the non, isn't the middle nonlinear,

C Charles Levine 27:36
okay,

A

Alex Gates 27:37

or is it monotonic, anyway, but the point is that maybe what we should be thinking about isn't the pools, and it's about how the pool interacts with your own ability, and then the nonlinearity is because you're adding people of lots of abilities together, right? So this is the other place where it could arise, these naturally, what

C

Charles Levine 27:58

you're saying,

A

Alex Gates 27:58

and it's not about how we assign people to teams. It's about how your team interacts with your own, and it's that term that needs to have some thought put into it.

C

Charles Levine 28:09

Yeah, right. No, I thought we saw this exact same thing when we were looking at the using survival analysis, when we were looking at the plots to show the the the risk of immediate promotion, it's a three letter acronym, I forgot what it was, right, the three dimensional plot,

A

Alex Gates 28:34

yep,

C

Charles Levine 28:34

the z axis was that value, and when you, when we, when we did the individual, leave one, leave everything else at the median, and just look at this. We got shapes that were completely different, but the conglomerate of them was a whole different ball game,

A

Alex Gates 28:52

right?

C Charles Levine 28:53
Remember what I'm talking about?

A Alex Gates 28:54
Yes, yes,

C Charles Levine 28:55
the hazard, the hazard ratio,

A Alex Gates 28:58
yeah,

C Charles Levine 28:58
the hazard ratio, the hazard ratios were completely in counterintuitive that they would come together to form the composite

A Alex Gates 29:09
right, so I almost want to say like this suggests looking at like to get this type of an effect happening

C Charles Levine 29:22
where we, the components, I'm just talking, yeah, we have the components of the high, middle, and low of different plots.

A

Alex Gates 29:29

Maybe it's something about your own ability minus the average team ability that matters, not the rank, but it's like, how much better are you than the type of person that you're being compared to, and this type of thing would give you actually quite naturally, maybe go after this first, forget these two, go after this first, the probability

S

Speaker 2 29:59

this.

C

Charles Levine 30:02

Isn't this the same as saying your ability, the standard or the deviation? Okay, just kind of like the standard. Can't we just use the standard deviation then of the of the pool

A

Alex Gates 30:18

with the rank, but with with rank we were saying, I don't care how much better you are than the rest of the pool, you're still number one. What I'm saying is what matters is how much better are you than the rest of the pool. How much better is it? How much better are you in comparison to your peers?

S

Speaker 2 30:37

Right,

A

Alex Gates 30:38

right, but again,

C

Charles Levine 30:39

this is that symbol there, the mean?

A Alex Gates 30:42
This is the, yeah, this is the mean of your pool,

S Speaker 2 30:44
that's expected value. Yeah, okay,

A Alex Gates 30:48
you could

C Charles Levine 30:50
do it, yeah, whatever you want. So we're saying, let's just take a look at the

A Alex Gates 30:55
sign of it matters, so you should get rid of that, right?

C Charles Levine 30:57
We'll just say, what's what's my, yeah, so if I have a positive number out of that, so what's my latent ability minus the,

A Alex Gates 31:05
yeah, I mean, you got to do something like e to the thing to make it all work in the way that you think it should,

C Charles Levine 31:12
yeah,

A

Alex Gates 31:13

right, but it should be something that says when I'm much better than the average of my pool, then I have a much higher chance of being distinction, distinction, but when I'm even if I'm the best person in my pool, if I'm very close to the typical performance of that pool,

S

Speaker 2 31:35

then

A

Alex Gates 31:36

I don't have a nearly as big of a benefit, right, that's the

C

Charles Levine 31:42

what if what if the variance in my pool is ginormous, like what if I have a very strong variance in my pool?

A

Alex Gates 31:52

Then the top person in that pool will be much better than the average of the pool. The variance is

C

Charles Levine 32:01

the right is

A

Alex Gates 32:02

the average of these values.

C

Charles Levine 32:04

You're right, right, you're right. Variance is the average. Thank you

A Alex Gates 32:11
for that

C Charles Levine 32:14
little obvious clue fairy that is fluid and landed on my shoulder. Okay, so you're saying, before we do anything else, let's just plot that. Let's just plot the relationship between a sub I minus the pool mean and probability of success It, but wait a minute, that's not going to give us these pool quantiles,

A Alex Gates 32:50
no. So, in the model that you build, you can integrate your own ability as part of it, right? You have a term, plus,

C Charles Levine 32:58
yeah, right,

A Alex Gates 32:59
plus your own,

C Charles Levine 32:59
you have a weight, you multiply the weight times the rank in the pool,

A Alex Gates 33:06
yeah,

C Charles Levine 33:06
and then add one minus the weight times the ability,

A Alex Gates 33:09
right. So I don't know if it should be your own ability or the difference between your ability and the team's1. of the terms in there has to be, let's

C Charles Levine 33:23
replace rank,

A Alex Gates 33:24
yeah, with this first,

C Charles Levine 33:26
that we say the weight times that,

A Alex Gates 33:29
yep,

C Charles Levine 33:30
plus one minus the weight times your actual a sub I, although algebraic,

A Alex Gates 33:36
that's just gonna cancel out,

S Speaker 2 33:38
yeah,

A Alex Gates 33:39
yeah, so I mean, that's that's actually what it will do, is it becomes now a weight, this minus
 λ your at minus one minus the weights kind of will multiply in,

C Charles Levine 33:57
so we've got w times I'm going to say $a_{I \text{ minus}}$

S Speaker 2 34:08
 $G_{\text{sub } J}$

C Charles Levine 34:12
plus one minus W times $A_{\text{sub } I}$,

A Alex Gates 34:18
right? And what that was simplified to with a slightly different interpretation of what W is $a_{\text{sub } I}$
minus W ,

C Charles Levine 34:28
right,

A Alex Gates 34:29
 $T_{\text{sub } I}$, that's what it'll simplify.

C Charles Levine 34:34
You have a negative $W A_{\text{sub } I}$. Here we have a positive $W A_{\text{sub } I}$, here,

A Alex Gates 34:40
yeah. And then you have a one,

S Speaker 2 34:42
right? Okay,

C Charles Levine 34:50
so it really, yeah,

S Speaker 2 34:54
nice.

A Alex Gates 34:56
So maybe try tweaking that rule a little bit,

C Charles Levine 34:59
yeah.

A Alex Gates 35:00
And whether it's this version exactly or something inspired by this idea,

C Charles Levine 35:05
right,

A

Alex Gates 35:05

that's the first thing to tweak, because the reason I said do that first is because actually articulating it, this is the easiest thing to do in the framework that you've already built, is literally just changing this term just slightly, right, the rest of it is adding new mechanisms and changing the actual sign. I

C

Charles Levine 35:25

really think this is necessary, but this is a, is it easy hanging fruit?

A

Alex Gates 35:29

Right. Okay, so do this first, because of that. I don't expect it to be a huge thing, just tweaking the code you have.

C

Charles Levine 35:36

Yeah,

A

Alex Gates 35:37

if that doesn't help at all, then move into this,

C

Charles Levine 35:40

yeah, and keep this in our back pocket for for fine tuning when we find what we need every year,

A

Alex Gates 35:47

yeah.

C Charles Levine 35:48
Okay, I can't absorb any more.

A Alex Gates 35:51
Great,

C Charles Levine 35:53
this is awesome,

A Alex Gates 35:54
but do you see how the all three suggestions focus on taking the rules we put into place and questioning those, not let's figure out new parameter combinations and do all this. So I want you to think more along those lines. How can you develop better rules in the model?

C Charles Levine 36:16
When you say rules, what do you mean by that?

A Alex Gates 36:18
Well, this is the actual rule of how we're assigning stuff. Oh, this is the rule of how promotion happens.

C Charles Levine 36:23

Okay, you're saying the algorithm for how we're building the algorithms we're building. Yeah, that's what I.. and I want you to know that's what I've been thinking about driving and stuff all week is reframe the way that we approach. Start with Scratch. I said the physicist in you. I'll never forget how pissed off I was. I was in physics class at the Citadel, and Peter Renbier was Polish, somebody said, "Well, he said, well, how do we.. what's the force of attraction between two masses, which I now know is not real, it's time space, but.. and somebody asked that. He's actually just give me the formula, r you know, man one and two over r squared. He said, "Well, let's do this. Let's take another mass out at infinity and slowly bring it in, and it's only in the last like 10 years of my life that I can appreciate that that's the way to, you know, say, yeah. And so that's the thing here, this, that's what I'm trying to do. I'm like, okay, let's start from scratch and intuitively build. Make sense?

A Alex Gates 37:18

Yep. Yeah. Good. Yes.

S Speaker 2 37:22

Great.

A Alex Gates 37:35

Thank